

Nazifa Mosharrat

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[LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

Professional Summary

Full stack developer with 5+ years of experience designing and developing scalable full-stack applications, backend systems, and business automation solutions across healthcare, logistics, finance, and enterprise domains. Strong hands-on background in Python, Java, C#, REST APIs, databases, cloud deployment, and containerized applications. Recently specialized in Artificial Intelligence through a Master's in Artificial Intelligence at University of Pisa, where academic and research projects focused on Generative AI, Retrieval-Augmented Generation (RAG), LLM applications, NLP, and machine learning systems. Built practical AI solutions including intelligent assistants, automated reporting tools, and domain-specific RAG systems using LangChain, Weaviate, and open-source LLMs. Combines proven industry software engineering experience with current advanced AI expertise, bringing the ability to translate emerging AI capabilities into reliable real-world applications.

Education

- Master's Degree in Artificial Intelligence | University of Pisa, Italy | 2023–2026 | 95/110
- BSc in Computer Science and Engineering | East Delta University, Bangladesh | 2019–2022 | GPA: 3.58/4.00
- Diploma in Computer Technology | Chittagong Mohila Polytechnic Institute, Bangladesh | 2013–2018 | CGPA: 3.84/4.00

Publication

Automatic Malware Categorization based on K-Means Clustering Technique

Publisher: Springer | DOI: https://doi.org/10.1007/978-981-16-6636-0_49 | September 2021 | 1st author

- Dynamically finds the optimum cluster number and feeds that cluster number into the K means clustering model.
- Categorized the malware family automatically into different batches
- Compared the proposed model with different existing systems using a real-time malware dataset

Undergraduate Thesis

Movie recommendation system based on watch time and genre-correlation

Supervisor: Sohrab Hossain

- The existing algorithm has been refined
- The recommender system has been improved by not only prioritizing the importance but also distribute the genre and watch time to find the correlation
- The most recommended movie from the dataset has been filtered out

Professional Experience

Research Assistant | January 2026– Present

- Conducting thesis research on advanced Graph RAG for the medical domain, with a specific focus on cardiology, using CUDA-accelerated computing, Weaviate, Qwen embedding models, and BGE-reranker-v2-m3 to improve retrieval accuracy, document reranking, and the reliability of generated responses.
- Engineered Smart KPI RAG system with LangChain, Ollama (LLaMA 3.2), and Weaviate, enabling real-time query response and automated report generation (PDF/email).
- Led MedPharm AI project using GPT-3.5 and RAG, resulting in enhanced supply chain optimization for medicine waste tracking via API orchestration.
- Implemented a CycleGAN-based unsupervised image denoising framework in PyTorch, analyzing distribution alignment and qualitative improvements under unpaired data settings.
- Built text-based emotion classification system using LSTM and BERT, enhancing multi-class NLP predictive accuracy.

- Created RNN model for benzene time-series forecasting, improving prediction performance with data-driven visualization.
- Designed multi-emission probabilistic modelling using HMMs, enabling high-precision state sequence analysis.

Software Engineer (Full Stack & System Integration)

Xceleran | Remote (USA) | Jul 2021 – Sep 2024

- Designed and developed scheduler management, charter service, POS, and virtual payment solutions.
- Built Pitch Deck analyser and vehicle tracking solution via Geotab API integration.
- Implemented Single Sign-On (SSO) across multiple business applications for seamless access.
- Collaborated cross-functionally to improve backend performance and system scalability.

Jr. Programmer (Full Stack Developer)

Neosysworld | Chattogram, Bangladesh | Nov 2020 – Aug 2021

- Developed travel management, ticket management for task allocation, courier logistics, inventory, and budgeting systems.
- Built “NSW Ticket” chatbot system (rule-based) for automated query response.
- Worked across database design, web development, and software deployment cycles.

Jr. Programmer – IT Department

Chattogram Maa-O-Shishu-O General Hospital | Chattogram, Bangladesh | Oct 2019 – Oct 2020

- Developed ERP modules: Pharmacy Management, HR System, and COVID-19 Test Booth automation.
- Designed database structures and automation workflows to support hospital operations.
- Assisted in software implementation and troubleshooting for clinical management systems.

Guest Instructor – Computer Technology Department

Chittagong Mohila Polytechnic Institute | Chattogram, Bangladesh | Feb 2019 – Sep 2019

Courses Taught: Computer fundamentals, Object Oriented Programming (JAVA), Database Management System, Web Mastering

Skills

- **Programming Languages:** Python, C#, Java
- **Machine Learning & AI Frameworks:** PyTorch, TensorFlow, Scikit-Learn, Transformers, Hugging Face, LangChain, GPT-3, Ollama 3.2, Stanza, BERT, LLaMA, GANs, CycleGen, StyleGAN, OpenCV, RAG (Retrieval-Augmented Generation)
- **Generative AI & LLM Technologies:** Fine-tuning LLMs, Prompt Engineering, RLHF (Reinforcement Learning from Human Feedback), Model Quantization (GGUF), LoRA, API Integration for AI Systems
- **Vector Databases & Knowledge Retrieval:** Weaviate (for RAG), FAISS, Qwen Embeddings, MaxBI embeddings, Qwen reranker, BGE-M3, BGE-reranker-v2-m3, semantic search, document reranking
- **Data Science & Analytics Tools:** Pandas, NumPy
- **Backend & API Development:** Flask, REST APIs, FastAPI, .NET Core, ASP.NET, Spring Boot
- **Frontend & Web Technologies:** AngularJS, Thymeleaf, HTML5, Razor, Blazor
- **GPU & Distributed Computing:** CUDA, GPU Computing, Distributed AI/ML Workloads
- **DevOps & MLOps:** Docker, AWS (EC2), GitHub, Bitbucket, Jira, CI/CD, Model Deployment & Optimization
- **Databases:** PostgreSQL, MySQL, Microsoft SQL Server
- **Cloud & Scalable AI Systems:** AWS, API-based Orchestration, Microservices
- **Integration & Business APIs:** OpenAI API, Google API, Payeezy, Quickbook, ZeoTab, Tua Pay, Wisetack

Certifications and Training

- **Generative AI with Diffusion Models** | NVIDIA Deep Learning Institute | Technical – Advanced Hands-on training on text-to-image generation using diffusion models and PyTorch, U-Net architecture, Forward/reverse diffusion, Denoising, CLIP for prompt-based image generation, Model optimization using GELU, GroupNorm, and sinusoidal embeddings.

- **Building RAG Agents with LLMs** | Self-Paced Course | Technical – Advanced
Hands-on training on deploying RAG-based agentic systems using large language models, implementing dialog management and internal reasoning, using embeddings for semantic retrieval and guardrail, designing pipelines with LangChain, Gradio, and LangServe, orchestrating document interactions, configuring vector stores for content retrieval, and evaluating modular RAG agents without fine-tuning.
- **Generative AI** | NVIDIA Deep Learning Institute | Technical – Beginner
Overview of Generative AI concepts and applications using neural networks.
- **Statistics for Data Science** | Great Learning | Issued Jan 2021
Fundamentals of statistical analysis, probability, hypothesis testing, regression, and data interpretation for data science.
- **Machine Learning with Python** | Great Learning | Issued Dec 2020
Introduction to supervised and unsupervised learning, model evaluation, feature engineering, and implementation in Python.
- **Advanced Game Development** | ICT Division (Government of Bangladesh) | Issued Sep 2016
Game architecture, logic development, gameplay programming, and performance optimization techniques.
- **Game Graphics Design** | ICT Division (Government of Bangladesh) | Issued Sep 2016
Visual design principles, 2D/3D asset creation, character modelling, UI/UX for games, and creative tools usage.

International English Language Test (IELTS)

Overall	Listening	Reading	Writing	Speaking
7.0	6.5	8.5	6.0	6.5